Control Statement (If else , switch ,for , while, break,etc.)

```
# if else with indentation error
a=23
b=302
if a>b:
    print("b is greater then a")
else:
print("a is greater then b")
```

```
  File "<ipython-input-1-990ba1a9f7f2>", line 7
    print("a is greater then b")
        ^
IndentationError: expected an indented block
```

```
# if else
a=23
b=302
if a>b:
    print("b is greater then a")
else:
    print("a is greater then b")
```

```
a is greater then b
```

```
msg = input("Are willing to learn Python 'yes' or 'no' : \n")
if msg=="yes":
    print("Welcome to Python Programming")
```

```
Are willing to learn Python 'yes' or 'no' :
yes
Welcome to Python Programming
```

```
#using only if
a=int(input("Enter no. 1 : "))
b=int(input("Enter no. 2 : "))
if a>b:
    print("no. 1 is greater then no. 2")
if a==b:
    print("no. 1 is equal to no. 2")
if b>a:
    print("no. 2 is greater then no. 1")
```

```
Enter no. 1 : 10
Enter no. 2 : 56
no. 2 is greater then no. 1
```

```
#using only if and else
a=int(input("Enter no. 1 : "))
b=int(input("Enter no. 2 : "))
if a>b:
    print("no. 1 is greater then no. 2")
if a==b:
    print("no. 1 is equal to no. 2")
else:
    print("no. 2 is greater then no. 1")
```

```
Enter no. 1 : 10
Enter no. 2 : 10
no. 1 is equal to no. 2
```

```
#using only if and else
a=int(input("Enter no : "))
if a>0:
    print("Positive")
if a==0:
    print("Zero")
else:
    print("Negative")
```

```
Enter no : -88
Negative
```

```
#using only if and else
a=int(input("Enter no : "))
if a%2==0 :
    print("Even")
else:
    print("Odd")
```

```
Enter no : 45646478313
Odd
```

```
#using only if , elif and else
a=int(input("Enter age : "))
if a>=18:
    print("Eligible to Vote")
elif a>0:
    print("Not eligible to Vote")
else:
    print("U are not a human !! Go away Demon !! (Splashing Holy water
on you)")
```

```
Enter age : -9999
U are not a human !! Go away Demon !! (Splashing Holy water on you)
```

```
a=int(input("Enter no. 1 : "))
b=int(input("Enter no. 2 : "))
```

```
c=int(input("Enter no. 3 : "))
if a>b and a>c:
    print(f"{a} is greatest")
elif a>c and b>a:
    print(f"{b} is greatest")
else:
    print(f"{c} is greatest")
```

```
Enter no. 1 : 10
Enter no. 2 : 50
Enter no. 3 : 13
13 is greatest
```

```
a=int(input("Enter no : "))
if a>0:
    print("Positive")
elif a==0:
    print("Zero")
else:
    print("Negative")
```

```
Enter no : -9
Negative
```

```
# nested if-else
a = int(input("Enter no. : "))
if a> 10:
    print("Above 10")
    if a>20:
        print("And also above 20")
    else :
        print("but not above 20")
```

```
Enter no. : 41
Above 10
And also above 20
```

```
# one line if-else
a=23
b=24
print("A") if a>b else print("=") if a==b else print("B")
```

```
B
```

Strings and string operations

```
a = "This is name1"
b = 'This is name2'
```

```
print(a)
print(b)
```

```
This is name1
This is name2
```

```
a = "This is 'name1'"
print(a)
b = 'This is "name2"'
print(b)
```

```
This is 'name1'
This is "name2"
```

```
# backslash character
a = "\"hello\" This isn\'t name1"
print(a)
```

```
"hello" This isn't name1
```

```
# accessing string character
print(a[0])
print(a[5])
print(a[-3])
```

```
"
o
m
```

```
# slicing
print(a[4:-11])
print(a[4:])
print(a[:-11])
print(a[-11:-4])
```

```
lo" This
lo" This isn't name1
"hello" This
isn't n
```

```
#concatenation
x = "Helllo"
y = "World"
z = "I"
a = "am"
b = "???"
print(a + b + y + z + x)
# string repition
print(3*(a + b + y + z + x))
```

```
am???WorldIHelllo
am???WorldIHellloam???WorldIHellloam???WorldIHelllo
```

```
# == operator
print("Hello" == "Hello")
print("Hello" == "hello")
```

```
True
False
```

```
# != operator
print("Hello" != "Hello")
print("Hello" != "hello")
```

```
False
True
```

```
# > operator
print("Hello" > "Hello")
print("Hello" > "hello")
```

```
False
False
```

```
# < operator
print("Hello" < "Hello")
print("Hello" < "hello")
```

```
False
True
```

```
print('jas' in 'jaspreet')
```

```
True
```

```
print('jas' not in 'jaspreet')
```

```
False
```

```
age = int(31)
print("age is %d"%(age))
```

```
age is 31
```

```
s='I am Jaspreet'
print(s[0:8])
```

```
I am Jas
```

```
s='abc'*3
print(s)
print(s[0:6:2])
print(s[0:6:2])
print(s[2:10:2])
print(s[::3])
print(s[1::3])
```

```
print(s[:8:3])

#3rd parameter as negative index(1st parameter then must be greater
then 2nd parameter)

print(s[8:1:-2])
print(s[8:1:-1])
print(s[8:-1:-1])
print(s[::-1])
```

```
abcabcabc
acb
acb
cbac
aaa
bbb
aaa
cabc
cbacbac

cbacbacba
abcabcabc
```